Dr. Snigdha Sen

Senior Assistant Professor, School of Computer Engineering
Manipal Institute of Technology Bengaluru, Manipal Academy of Higher Education (MAHE)

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RESEARCH INTERESTS

Astroinformatics and astronomical big data; deep learning for photometric redshift estimation; explainable AI for space science; computer vision and image steganography; scalable machine learning on big-data platforms.

PROFILE AT A GLANCE

- About 16 years of teaching experience in computer science and engineering; PhD (2024) from IIIT Allahabad.
- 57 publications, 46 of them indexed in Scopus (260 citations, h-index 9, as of Oct 2026), including *New Astronomy*, *Experimental Astronomy*, *Signal, Image and Video Processing* and *Multimedia Tools and Applications*.
- Two patents in emergency-response systems; funded projects from VGST and KSCST.
- Guides 4 doctoral scholars at MAHE; has supervised 10+ M.Tech theses and 35+ undergraduate projects.
- Invited speaker at IUCAA Pune, ISRO ISTRAC, IIIT Allahabad and IIIT Lucknow.
- Editorial board member: *The Visual Computer*; *Frontiers in Big Data*.

ACADEMIC APPOINTMENTS

Senior Assistant Professor — Mar 2024 – present
Manipal Institute of Technology Bengaluru, MAHE
School of Computer Engineering; Research Coordinator.

Associate Professor — Aug 2011 – Feb 2024
Global Academy of Technology, Bengaluru
Organised conferences, workshops and faculty development programmes; resource person for ML, deep learning, big data and cloud computing.

Lecturer, Computer Science — Jul 2005 – Jul 2007
Aryabhatta Institute of Engineering & Management, Durgapur

Industry

Programmer Analyst Trainee — Feb 2010 – Jun 2010
Cognizant Technology Solutions, Kolkata

EDUCATION

Ph.D., Data Science & Deep Learning — 2019 – 2024
Indian Institute of Information Technology Allahabad
Supervisors: Prof. Pavan Chakraborty and Dr. Krishna Pratap Singh.

M.Tech., Information Technology — 2007 – 2009
Jadavpur University, Kolkata

B.E., Information Technology — 2001 – 2005
University Institute of Technology, University of Burdwan

PUBLICATIONS

Reverse chronological. Indexing shown where known. Full list with links: snigdhasen.in/publications

Journal Articles

[1] A. Dabhade, S. Chakraborty, and **S. Sen**, "A novel block based preparation network using convolutional neural networks for image steganography and robust data protection," *Multimedia Tools and Applications*, vol. 85, no. 5, art. 482, 2026. doi:10.1007/s11042-026-21652-5 [Scopus]

[2] R. DSouza and **S. Sen**, "A SwinBERT framework with temporal windowed cross attention for video captioning for visual language intelligence," *Discover Applied Sciences*, vol. 8, no. 9, art. 947, 2026. doi:10.1007/s42452-026-09098-8 [Scopus]

[3] S. Nithya, S. Banerjee, **S. Sen**, and S. Senthil Kumar, "Explainable Deep Learning for Satellite Telemetry Anomaly Detection: A Systematic Literature Review," *IEEE Access*, 2026. doi:10.1109/ACCESS.2026.3745984 [SCIE, Scopus]

[4] D. Mahesh, N. V. Raju, A. Pasha, and **S. Sen**, "Prediction of PM2.5 Using Machine Learning Algorithms: A Case Study for Bengaluru, India," *Aerosol Science and Engineering*, 2026 (accepted, in press). [Scopus]

[5] A. Sk, S. Chakraborty, and **S. Sen**, "A novel blocked based U-Net model for image steganography," *Multimedia Tools and Applications*, vol. 84, no. 36, pp. 45479–45498, 2025. doi:10.1007/s11042-025-20952-6 [Scopus]

[6] S. Madhavi, D. Mahesh, N. V. Raju, **S. Sen**, and J. Johns, "Analysing and predicting dielectric properties of bamboo fiber filled polypropylene composites using artificial neural network," *Journal of the Indian Academy of Wood Science*, vol. 22, no. 1, pp. 37–49, 2025. doi:10.1007/s13196-024-00358-x [Scopus]

[7] D. Mahesh, N. V. Raju, and **S. Sen**, "Machine learning approaches for predicting dielectric properties of banana fibers reinforced with polypropylene composites," *Multiscale and Multidisciplinary Modeling, Experiments and Design*, vol. 8, no. 1, art. 71, 2025. doi:10.1007/s41939-024-00666-2 [Scopus]

[8] **S. Sen**, A. K. Pandit, P. Chakraborty, and K. P. Singh, "Performance evaluation of efficient interpretable CNN-transformer model for redshift prediction," *Signal, Image and Video Processing*, vol. 19, no. 4, art. 326, 2025. doi:10.1007/s11760-025-03855-9 [SCI, Scopus]

[9] **S. Sen** and S. Banerjee, "Precise Image Segmentation using Explainable U-Net based Hybrid Approach for Improved Detection of Skin Lesion," *International Journal of Computing and Digital Systems*, vol. 18, no. 1, pp. 1–12, 2025. doi:10.12785/ijcds/1571149929 [Scopus]

[10] M. Doddashamachar, **S. Sen**, and R. Nama Vasudeva Setty, "A novel Artificial Neural Network-based model for predicting dielectric properties of banana fiber filled with polypropylene composites," *Journal of Thermoplastic Composite Materials*, vol. 36, no. 10, pp. 4106–4123, 2023. doi:10.1177/08927057221148455 [SCI, Scopus]

[11] **S. Sen**, K. P. Singh, and P. Chakraborty, "Dealing with imbalanced regression problem for large dataset using scalable Artificial Neural Network," *New Astronomy*, vol. 99, art. 101959, 2023. doi:10.1016/j.newast.2022.101959 [SCI, Scopus]

[12] **S. Sen**, S. Agarwal, P. Chakraborty, and K. P. Singh, "Astronomical big data processing using machine learning: A comprehensive review," *Experimental Astronomy*, vol. 53, no. 1, pp. 1–43, 2022. doi:10.1007/s10686-021-09827-4 [SCI, Scopus]

[13] **S. Sen**, "Comparative Study of Methodologies for Home Automation System: A Survey," *International Journal of Computer Sciences and Engineering (IJCSE)*, 2019.

[14] **S. Sen**, "Development of an IoT Based Shopping Trolley," *International Journal on Recent and Innovation Trends in Computing and Communication (IJRITCC)*, 2019.

[15] **S. Sen**, "Development of an Exclusive Web Application for Family," *International Journal of Advanced Research in Computer Science*, 2018.

[16] **S. Sen**, "Implementation of deduplication scheme for Encrypted Data in Cloud," *International Journal of Innovative Research in Computer and Communication Engineering (IJIRCCE)*, 2017.

[17] **S. Sen** and B. Madhu, "Smart agriculture: a bliss to farmers," *International Journal of Engineering Sciences & Research Technology*, vol. 6, no. 4, pp. 197–202, 2017.

Conference Proceedings

[18] H. Alla, **S. Sen**, and J. Saha, "3D CNNs for Hyperspectral-Based Water Potability Detection," in *Soft Computing and Its Engineering Applications (icSoftComp 2025)*, Communications in Computer and Information Science, vol. 2873, pp. 208–221, Springer Nature Switzerland, 2026. doi:10.1007/978-3-032-22059-2_16 [Scopus]

[19] S. S. Parvez, R. Kumar, M. D. Shaheer, **S. Sen**, and P. Chakraborty, "Efficient LLM Framework Using LORA Based Fine Tuning to Analyze Textual Astronomical Data," in *Soft Computing and Its Engineering Applications (icSoftComp 2025)*, Communications in Computer and Information Science, vol. 2874, pp. 16–28, Springer Nature Switzerland, 2026. doi:10.1007/978-3-032-22062-2_2 [Scopus]

[20] A. Das, M. Satyanarayana, **S. Sen**, and M. Shanthi, "Explainable Neural Networks for Interpretable Biometric Data Analysis," *Procedia Computer Science*, vol. 283, pp. 5412–5422, 2026. doi:10.1016/j.procs.2026.06.591 [Scopus]

[21] R. Tufani, **S. Sen**, and P. Chakraborty, "Comprehensive Performance Analysis of PySpark and Pandas for Classification and Clustering Task," in *Advances in Health Informatics, Intelligent Systems, and Networking Technologies*, Lecture Notes in Networks and Systems, pp. 251–262, Springer Nature Singapore, 2025. doi:10.1007/978-981-96-4008-9_19 [Scopus]

[22] J. Saha, M. Kumar, A. Shah, and **S. Sen**, "Extensive Log Analysis Using Small Language Models in Minimal Resource Environments," *2025 IEEE International Conference on Automatic Control and Intelligent Systems (I2CACIS)*, pp. 500–505, IEEE, 2025. doi:10.1109/I2CACIS65476.2025.11100950 [Scopus]

[23] R. Tufani, **S. Sen**, and P. Chakraborty, "Harnessing The Potential of Pandas and PySpark: An In-Depth Performance Evaluation for Machine Learning Classification on Single Node Architectures," *2025 International Conference on Artificial Intelligence and Machine Vision (AIMV)*, pp. 1–6, IEEE, 2025. doi:10.1109/AIMV66517.2025.11203526 [Scopus]

[24] A. Hassan, A. Juturu, **S. Sen**, and R. Dsouza, "Real-Time Object Detection and Tracking Using Mobinet and SSD Algorithms," in *Intelligent Systems (International Conference on Machine Learning, IoT and Big Data)*, Lecture Notes in Networks and Systems, pp. 449–462, Springer Nature Singapore, 2025. doi:10.1007/978-981-96-3797-3_36 [Scopus]

[25] **S. Sen** and P. Chakraborty, "Evaluating Efficacy of MAML Based Approach on Regression Using Astronomical Imbalanced Dataset," *2024 International Conference on Computational Intelligence and Network Systems (CINS)*, pp. 1–8, IEEE, 2024. doi:10.1109/CINS63881.2024.10864404 [Scopus]

[26] M. Manasa and **S. Sen**, "Interpretable Predictions for crime categories using log loss approach for imbalanced target feature," *2024 IEEE International Conference on Contemporary Computing and Communications (InC4)*, pp. 1–6, IEEE, 2024. doi:10.1109/InC460750.2024.10649054 [Scopus]

[27] **S. Sen**, P. Chakraborty, S. Das, K. Pandey, and P. Narayana, "Investigation of Clustering Methods for SDSS Galaxy Images through Feature Extraction with VGG-16," *2024 IEEE Space, Aerospace and Defence Conference (SPACE)*, pp. 660–664, IEEE, 2024. doi:10.1109/SPACE63117.2024.10667753 [Scopus]

[28] D. S. Sahana, K. S. Varsha, **S. Sen**, and R. Priyanka, "A CNN-Based Approach for Facial Emotion Detection," in *Soft Computing: Theories and Applications - Proceedings of SoCTA 2022*, Lecture Notes in Networks and Systems, vol. 627, pp. 1–10, Springer Science and Business Media, 2023. doi:10.1007/978-981-19-9858-4_1 [Scopus]

[29] **S. Sen**, S. Saha, P. Chakraborty, and K. P. Singh, "A Fast and Robust Photometric Redshift Forecasting Method Using Lipschitz Adaptive Learning Rate," in *Neural Information Processing (ICONIP 2022)*, Communications in Computer and Information Science, pp. 123–135, Springer Nature Singapore, 2023. doi:10.1007/978-981-99-1642-9_11 [Scopus]

[30] K. Shreevershith, **S. Sen**, and G. B. Roopesh, "An Energy-Efficient Deep Neural Network Model for Photometric Redshift Estimation," in *Innovations in Computer Science and Engineering*, Lecture Notes in Networks and Systems, pp. 319–330, Springer Nature Singapore, 2023. doi:10.1007/978-981-19-7455-7_24 [Scopus]

[31] J. Parnika, **S. Sen**, R. S. Divya, S. S. Deepika, and V. Dharshini, "An Explainable Machine Learning Model to Analyze and Detect Fake Currency," *2023 International Conference on Communication, Circuits, and Systems, IC3S 2023*, pp. 1–6, IEEE, 2023. doi:10.1109/ic3s57698.2023.10169455 [Scopus]

[32] M. Varsha, Y. R. Aithal, S. A. Fathima, and **S. Sen**, "Distress Detection Using a Hybrid SVM - CNN Classifier," *Proceedings - 2023 International Conference on Computational Intelligence for Information, Security and Communication Applications, CIISCA 2023*, pp. 224–229, IEEE, 2023. doi:10.1109/ciisca59740.2023.00051 [Scopus]

[33] **S. Sen**, V. T. Rajashekar, and N. Dharshan, "Forecasting and Analysing Time Series Data Using Deep Learning," in *Intelligent Systems - Proceedings of 3rd International Conference on Machine Learning, IoT and Big Data ICMIB 2023*, Lecture Notes in Networks and Systems, vol. 728, pp. 279–291, Springer Science and Business Media, 2023. doi:10.1007/978-981-99-3932-9_25 [Scopus]

[34] M. I. Hegde, M. P. M. Shivalingappa, and **S. Sen**, "Paramount—A Hidden Markov Model Based Intelligent Voice Assistant," *Proceedings of International Conference on Data Analytics and Insights, ICDAI 2023*, Lecture Notes in Networks and Systems, vol. 727, pp. 755–765, Springer Science and Business Media, 2023. doi:10.1007/978-981-99-3878-0_64 [Scopus]

[35] A. Kumar, M. K. Priyanshu, R. Singh, and **S. Sen**, "Plant Leaf Disease Detection Using ResNet," in *Soft Computing: Theories and Applications - Proceedings of SoCTA 2022*, Lecture Notes in Networks and Systems, vol. 627, pp. 867–877, Springer Science and Business Media, 2023. doi:10.1007/978-981-19-9858-4_74 [Scopus]

[36] N. Dhanvina, N. D. Mahesh, and **S. Sen**, "Speech-enabled machine learning-based automated attendance monitoring system through face recognition," *Proceedings of IEEE InC4 2023 - 2023 IEEE International Conference on Contemporary Computing and Communications*, pp. 1–6, IEEE, 2023. doi:10.1109/inc457730.2023.10263205 [Scopus]

[37] K. M. Bhargav, A. Bhat, **S. Sen**, A. V. K. Reddy, and S. D. Ashrith, "Voice-Based Intelligent Virtual Assistant for Windows," in *Innovations in Computer Science and Engineering - Proceedings of the 10th ICICSE, 2022*, Lecture Notes in Networks and Systems, vol. 565, pp. 491–500, Springer Science and Business Media, 2023. doi:10.1007/978-981-19-7455-7_36 [Scopus]

[38] Pankaj, **S. Sen**, and P. Chakraborty, "A Novel Classification-Based Approach for Quicker Prediction of Redshift Using Apache Spark," *2022 International Conference on Data Science, Agents and Artificial Intelligence, ICDSAAI 2022*, pp. 1–6, IEEE, 2022. doi:10.1109/icdsaai55433.2022.10028971 [Scopus]

[39] R. Monisha, **S. Sen**, R. U. Davangeri, K. S. S. Lakshmi, and S. Dey, "An Approach Toward Design and Implementation of Distributed Framework for Astronomical Big Data Processing," in *Intelligent Systems - Proceedings of ICMIB 2021*, Lecture Notes in Networks and Systems, vol. 431, pp. 267–275, Springer Science and Business Media, 2022. doi:10.1007/978-981-19-0901-6_26 [Scopus]

[40] I. Amrita, **S. Sen**, and K. Ashwini, "Analysis of Trend in COVID-19, World Vaccination and Its Side Effects Using Machine Learning," in *Advances in Distributed Computing and Machine Learning - Proceedings of ICADCML 2022*, Lecture Notes in Networks and Systems, vol. 427, pp. 317–326, Springer Science and Business Media, 2022. doi:10.1007/978-981-19-1018-0_27 [Scopus]

[41] N. S. Khasnis, S. S. Khasnis, and **S. Sen**, "Analytics to Preserve the Street Vendor Tradition of India and to Optimize profits post Covid-19," *2022 1st International Conference on Computational Science and Technology, ICCST 2022 - Proceedings*, pp. 1–6, IEEE, 2022. doi:10.1109/iccst55948.2022.10040318 [Scopus]

[42] N. S. Khasnis, S. S. Khasnis, and **S. Sen**, "Care-In-A-Cup: A Solution to Tend to the Needs of Autistic and Differently Abled Women," *2022 IEEE International Conference for Women in Innovation, Technology and Entrepreneurship, ICWITE 2022 - Proceedings*, pp. 1–6, IEEE, 2022. doi:10.1109/icwite57052.2022.10176224 [Scopus]

[43] K. Mayank, **S. Sen**, and P. Chakraborty, "Implementation of Cascade Learning using Apache Spark," *2022 IEEE International Conference on Electronics, Computing and Communication Technologies, CONECCT 2022*, pp. 1–6, IEEE, 2022. doi:10.1109/conecct55679.2022.9865798 [Scopus]

[44] N. S. Khasnis, **S. Sen**, and S. S. Khasnis, "A Machine Learning Approach for Sentiment Analysis to Nurture Mental Health Amidst COVID-19," *Proceedings of the International Conference on Data Science, Machine Learning and Artificial Intelligence, DSMLAI 2021*, pp. 284–289, Association for Computing Machinery, 2021. doi:10.1145/3484824.3484877 [Scopus]

[45] **S. Sen** and I. Amrita, "A Transfer Learning Based Approach for Lung Inflammation Detection," in *Advanced Techniques for IoT Applications - Proceedings of EAIT 2020*, Lecture Notes in Networks and Systems, vol. 292, pp. 29–36, Springer Science and Business Media, 2021. doi:10.1007/978-981-16-4435-1_4 [Scopus]

[46] S. VY, **S. Sen**, and K. Santosh, "Analyzing and Processing of Astronomical Images using Deep Learning Techniques," *Proceedings of CONECCT 2021: 7th IEEE International Conference on Electronics, Computing and Communication Technologies*, pp. 01–06, IEEE, 2021. doi:10.1109/conecct52877.2021.9622583 [Scopus]

[47] **S. Sen**, S. Saha, P. Chakraborty, and K. P. Singh, "Implementation of Neural Network Regression Model for Faster Redshift Analysis on Cloud-Based Spark Platform," in *Advances and Trends in Artificial Intelligence. From Theory to Practice (IEA/AIE 2021)*, Lecture Notes in Computer Science, pp. 591–602, Springer International Publishing, 2021. doi:10.1007/978-3-030-79463-7_50 [Scopus]

[48] **S. Sen**, "Analysis, Visualization and prediction of Covid19 pandemic spread using Machine Learning," *LNCS, Springer*, 2020. [Scopus]

[49] **S. Sen**, "Internet of Things: an Introduction to Connecting the Unconnected," in *International Journal of Engineering Research & Technology (IJERT)*, 2016.

Book Chapters

[50] B. Madhu and **S. Sen**, "An AI-Based Secure and Robust Customized Patient Healthcare System Using Blockchain Technology," in *AI in Smart and Secure Healthcare*, Springer Optimization and Its Applications, pp. 449–468, Springer Nature Switzerland, 2026. doi:10.1007/978-3-032-15092-9_18 [Scopus]

[51] M. I. Hegde, N. Dhanvina, N. Dharshan, and **S. Sen**, "Investigation of Machine Learning Algorithms in Detecting Chronic Kidney Disorder," in *Machine Learning for Disease Detection, Prediction, and Diagnosis*, pp. 207–224, Springer Nature Singapore, 2025. doi:10.1007/978-981-96-4241-0_11 [Scopus]

[52] **S. Sen** and B. Madhu, "Revolutionizing Quantum Cryptography With Artificial Intelligence: A Perspective for Collaborative Research," in *Advancing Cyber Security Through Quantum Cryptography*, Advances in Information Security, Privacy, and Ethics, pp. 441–464, IGI Global, 2024. doi:10.4018/979-8-3693-5961-7.ch016

[53] I. Amrita and **S. Sen**, "Deep Learning Algorithms for Classification and Prediction of Acute Lymphoblastic Leukemia," in *Current Applications of Deep Learning in Cancer Diagnostics*, pp. 115–124, CRC Press, 2023. doi:10.1201/9781003277002-9 [Scopus]

[54] **S. Sen** and Y. R. Aithal, "Evaluation of deep learning models for identification and prediction of new strains of COVID-19," in *Computer Intelligence Against Pandemics: Tools and Methods to Face New Strains of COVID-19*, pp. 257–286, de Gruyter, 2023. doi:10.1515/9783110767681-013 [Scopus]

Selected Articles in CSI Communications

[55] **S. Sen**, "An Overview of Semantic Web," *CSI Communications*, 2020.

[56] **S. Sen**, "Quantum Computing: A new phase of Computing," *CSI Communications*, 2020.

[57] **S. Sen**, "Robotics as learning tutor and its aid in education," *CSI Communications*, 2020.

RESEARCH GRANTS

Project	Role	Agency	Period
Predicting the likelihood of PCOD from physical and psychological factors in young adult women using machine learning (₹3 lakh)	Co-PI	VGST, Govt. of Karnataka	2019–20
Constructing small language models as subject-matter experts using minimal resources (student project grant)	Mentor / Guide	KSCST	2023–24
Care-In-A-Cup: a solution to tend to the needs of differently abled women (student project grant)	Mentor / Guide	KSCST	2021–22

PATENTS

- A system and method for multimodal on-device emergency detection with indoor localization and alerts. *Published.*
- A system and method for optimizing emergency medical response through real-time data integration. *Published.*

Proposals under review

- Investigation of the Baby Dragon Hatchling (BDH) architecture for satellite remote sensing image applications — ISRO RESPOND.
- A data-driven and sustainable AI/ML solution for enhancing rural education in Karnataka — VGST.
- Digital preservation of Indian ancient heritage using AI and immersive technologies: a metaverse approach.

RESEARCH SUPERVISION

- **Ph.D.:** guiding 4 doctoral scholars at MAHE (ongoing).
- **M.Tech.:** 10+ theses supervised.
- **B.E./B.Tech.:** 35+ projects supervised, including state-level exhibition and national ideathon winners.

INVITED TALKS

- Transformer-based models for astronomical data — guest talk, **IUCAA, Pune**.
- AI/ML training programme — **ISRO Telemetry, Tracking and Command Network (ISTRAC)**.
- Skewed data distributions — guest lecture, **IIIT Allahabad**.
- Deep Learning & its Application — invited talk, **IIIT Lucknow**.

PROFESSIONAL SERVICE

- Editorial board member, *The Visual Computer* and *Frontiers in Big Data*.
- Session chair at international conferences in India and abroad.
- Lead editorial member for two forthcoming edited books.
- Organiser and resource person for conferences, workshops and faculty development programmes.
- Life member, Computer Society of India (CSI); organised CSI student-chapter events.

AWARDS AND RECOGNITION (WITH STUDENT TEAMS)

Winner, National-level Ideathon Jun 2021
VVCE IEEE Computer Society Student Chapter
Web app using data analytics to preserve the street-vendor tradition of India post-COVID-19.

Winner, Paper Presentation Dec 2020
34th CSI Karnataka State Student Convention, KSIT Bengaluru
"Working with Docker: The New Industry Need."

Selected for State-level Exhibition
KSCST and VTU
"Care-In-A-Cup: A Solution to Tend to the Needs of Differently Abled Women."